Harness

File: harness.kicad_sch

Daughterboard

File: daughter.kicad_sch

LSU, MAP, IAT, USB, AUX

File: io.kicad_sch

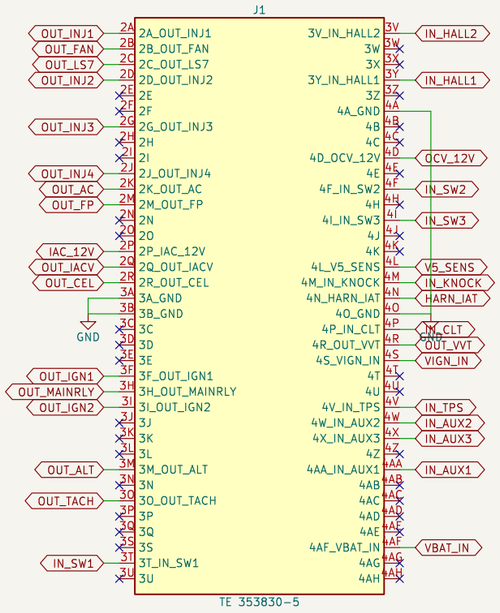
Miata ECU motherboard 2001–2005 (NB2, VVT). Carries the year-specific harness header and the year-independent I/O (wideband, MAP, IAT, USB-C). All engine control is on the daughterboard.

Sheet: /
File: miata_mb_0105.kicad_sch

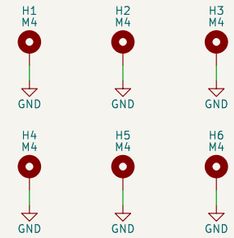
**Title: Miata ECU motherboard 2001–2005 (NB2, VVT)**

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-29 | **Rev: 0.2** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/4 |

## Stock harness header (Miata ECU motherboard 2001–2005): every pin labelled with the daughterboard signal it drives




## Case mounting bosses (M4), tied to ground




Generated from scripts/mb_nb2.py; netlist identical to the routed board

Sheet: /Harness/
File: harness.kicad_sch

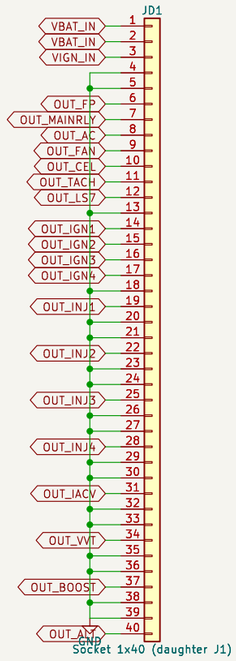
**Title: Miata ECU motherboard 2001–2005 — Harness**

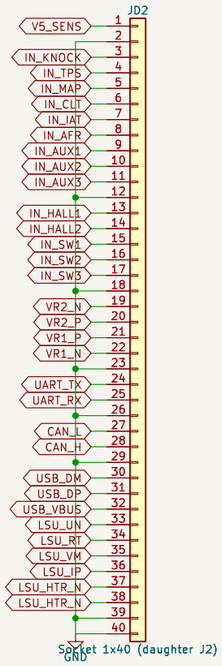
| Size: A3 | Date: 2026-10-29 | **Rev: 0.2** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/4 |

## Daughterboard sockets: pin-for-pin with the daughterboard J1 (top) and J2 (bottom)

## Daughterboard standoffs (M3)

HD1 M3 standoff
HD2 M3 standoff
HD3 M3 standoff
HD4 M3 standoff




Generated from scripts/mb_nb2.py; netlist identical to the routed board

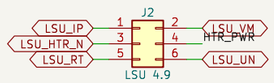
Sheet: /Daughterboard/
File: daughter.kicad_sch

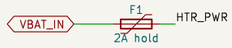
**Title: Miata ECU motherboard 2001–2005 — Daughterboard**

| Size: A3 | Date: 2026-10-29 | **Rev: 0.2** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/4 |

## LSU 4.9 wideband sensor (controller on the daughterboard)

LSU 4.9: 1 IP  2 VM  3 H−  4 H+  5 RT  6 UN

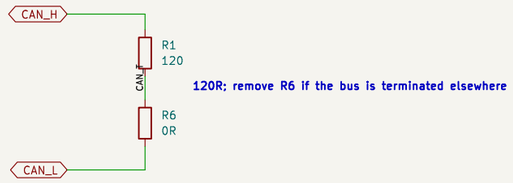



Heater fuse: 2 A hold, 30 V PTC

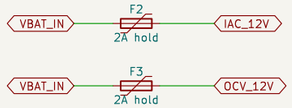
IP/VM/RT/UN and H− run to daughterboard J2 33–38; the heater current returns on J2 39/40

## CAN bus termination at the ECU end

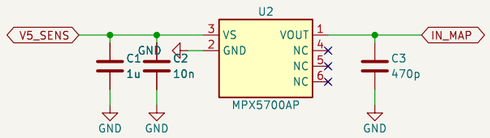



120R; remove R6 if the bus is terminated elsewhere

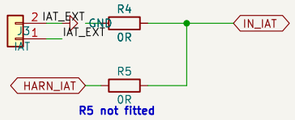
## Fused +12 V feeds to harness actuators




## MAP sensor

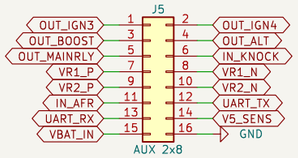



## IAT: onboard connector (R4, default) or stock harness (R5)

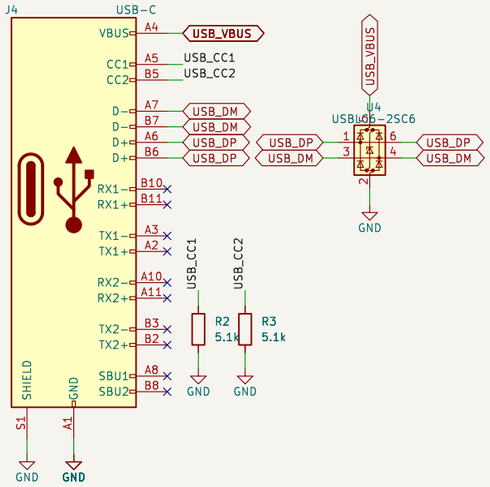



R5 not fitted

## AUX: add−on I/O the stock harness has no pins for




## USB−C (device): tuning, logs, DFU firmware update




Generated from scripts/mb_nb2.py; netlist identical to the routed board

Sheet: /LSU, MAP, IAT, USB, AUX/
File: io.kicad_sch

**Title: Miata ECU motherboard 2001–2005 — LSU, MAP, IAT, USB, AUX**

| Size: A3 | Date: 2026−10−29 | **Rev: 0.2** |
|---|---|---|
| KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4 | | Id: 4/4 |